

MAT235: Vector Calculus I

Week 1

September 7, 2026

Instructor Introduction

Dr. Yi Shan

Postdoctoral fellow

Pronouns: she/her

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You can call me: Yi (or Dr. Shan if you really want to be formal)

Other things about me:

PhD from École Normale Supérieure in 2025

Joined UofT Fall 2025

Research interests: Automorphic forms, representation theory, Langlands program

Personal interests: Reading, listening to metal music.



MAT235 at a glance

MAT235 is essentially a continuation of the differentiation content from MAT135/137. The course is organized around the following topics:

- 1 Functions: multivariable functions and vector fields.

Vector Calculus II (MAT236) continues this material in the Winter semester, and focuses on the integral calculus of several variables.

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- 3 Applications: critical points and optimization.

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- Evaluation and assessments: **read the syllabus!**

Recall: one-variable functions

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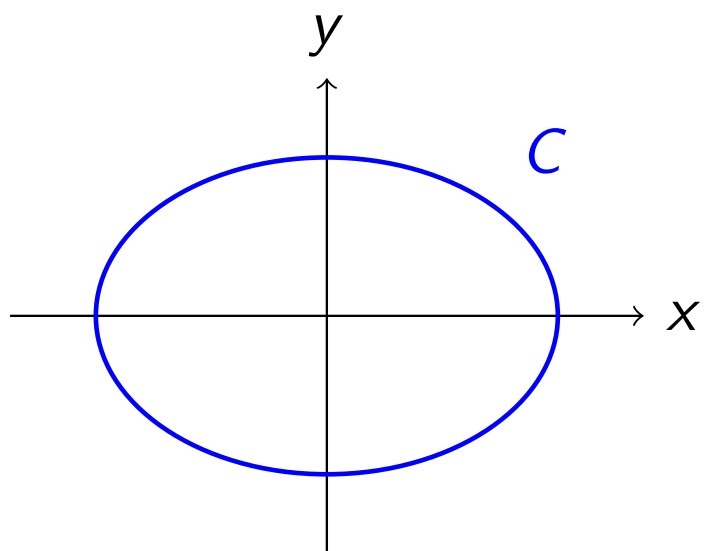
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- 3 numerically, by a table of values;
- 4 verbally.

Is it a function?



Is C the graph of a function $y = f(x)$?
Why or why not?

Function notation

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Definition (Two variable function)

The notation $f(x, y)$ denotes the value of the function f with inputs x and y . The inputs are the *independent variables*; a variable representing the resulting value is the *dependent variable*.

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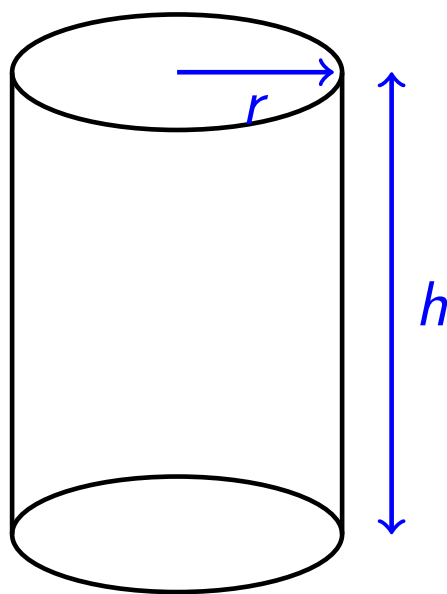
A function of two variables can be described in four ways:

- 1 algebraically, by a formula;
- 2 graphically, using graphs, charts, or contour diagrams;
- 3 numerically, by a table of values;
- 4 verbally.

Algebraic example: Volume and surface of a cylinder

Example

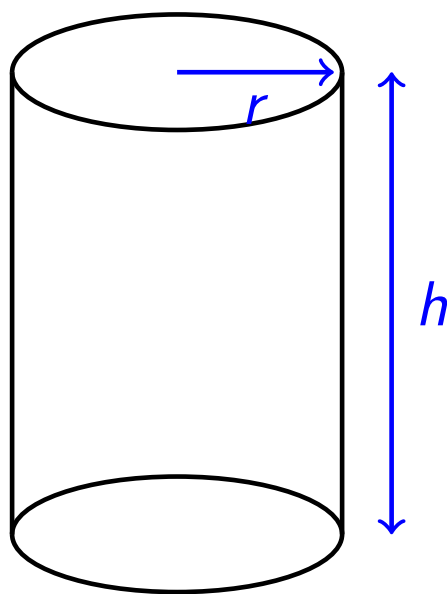
A cylinder with closed ends has radius r and height h . Its volume is V and its surface area is A . Find formulas for $V = f(r, h)$ and $A = g(r, h)$.



Algebraic example: Volume and surface of a cylinder

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Which variables are independent variables? Which are dependent variables?

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$w \backslash T$	35	30	25	20	15	10	5	0
5	31	25	19	13	7	1	-5	-11
10	27	21	15	9	3	-4	-10	-16
15	25	19	13	6	0	-7	-13	-19
20	24	17	11	4	-2	-9	-15	-22
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- 1 Evaluate and interpret $f(20, 5)$.
- 2 How fast does the wind need to blow for it to feel like -10°F when the air temperature is really 5°F ?

Graphical example: Increasing and decreasing behavior

Let $T = f(d, t)$ be the temperature, in degrees Fahrenheit, of a room t minutes after a heater is turned on, at a distance of d meters from the heater.

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- 1 Is T an increasing or a decreasing function of t ? Explain. Sketch possible cross sections of f for $d = 1, 2$ and 3 meters.
- 2 Is T an increasing or a decreasing function of d ? Explain. Sketch possible cross sections of f for $t = 0, 5$ and 10 minutes.

- About Quiz 1: (§12.1-12.2)

Quizzes - Pages - Quiz 1 information

- Week 2's pre-class reading:

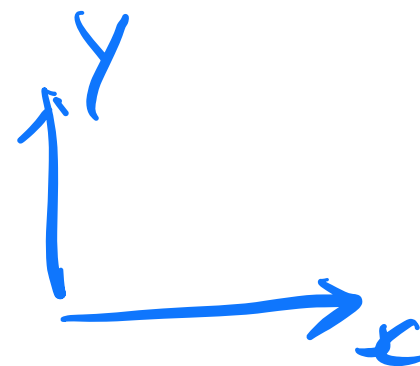
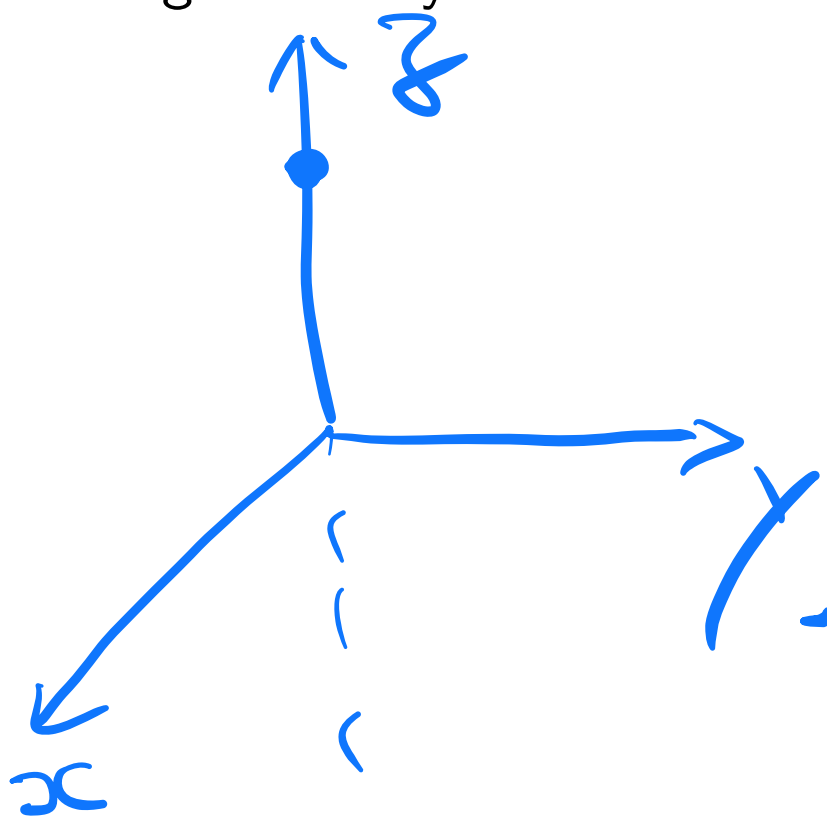
Quizzes - Pages - Week 2 Pre-class reading

(I'm confused; it's written that
" should be completed before Sept. 8 too)

Worksheet (upload before next Monday)

A quick tour of 3-space

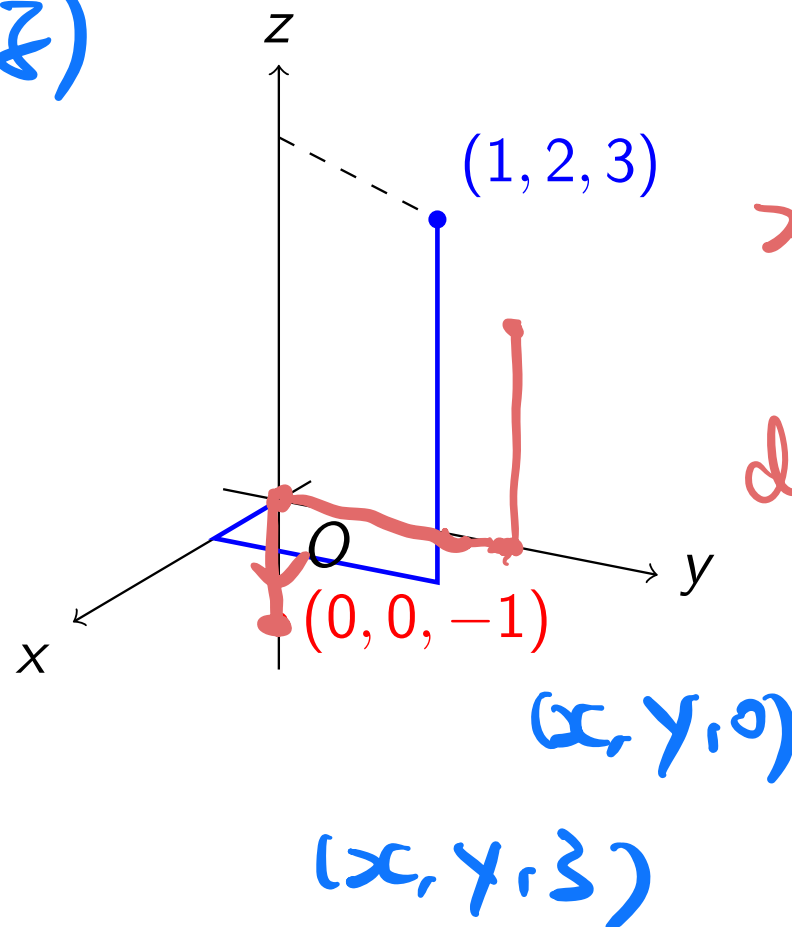
We generally use *right-handed axes* (looking down the positive z -axis gives the usual view of the xy -plane), with the xy -plane horizontal and the z -axis extending vertically above and below the plane.



A quick tour of 3-space

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(x, y, z)



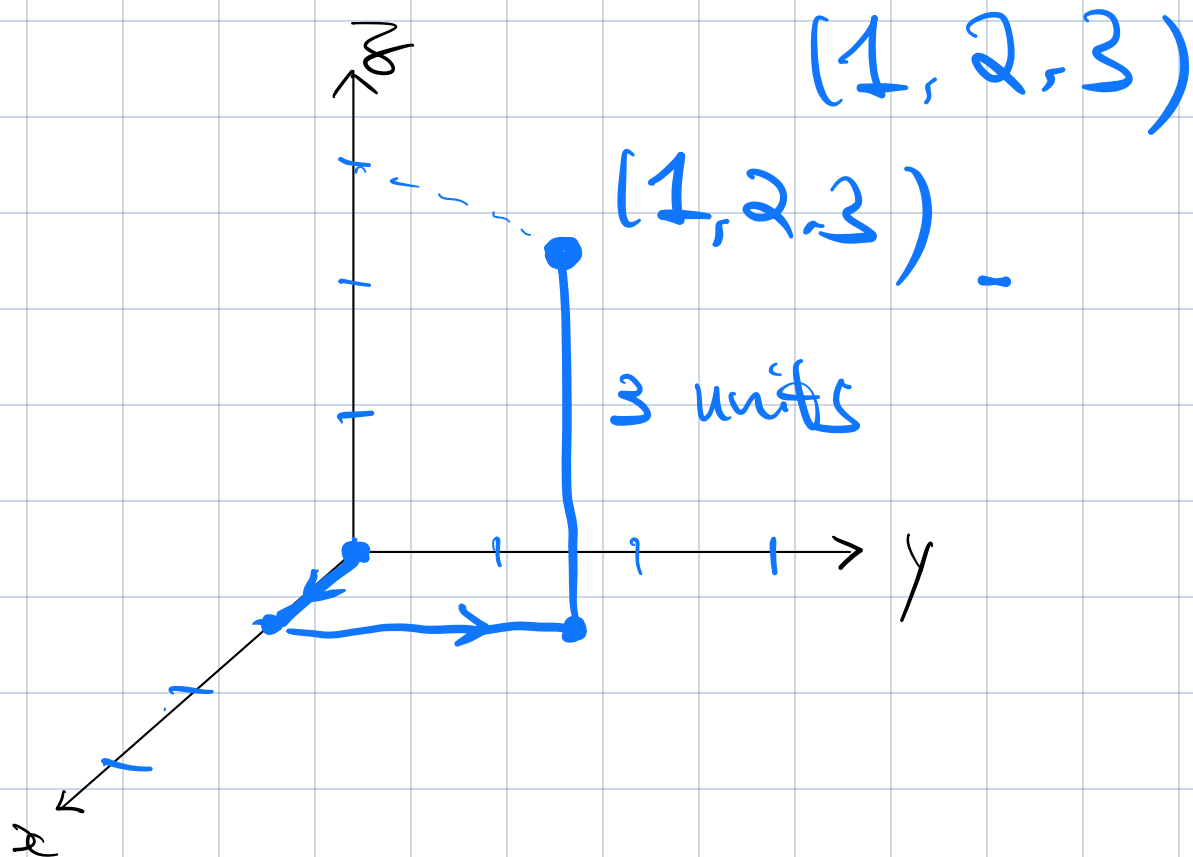
Viewing the coordinates (x, y, z) as instructions telling you how to get to the point:

- 1 Start at the origin;
- 2 move x units along the x -axis;
- 3 move y units parallel to the y -axis;
- 4 move z units parallel to the z -axis.

$x=0$

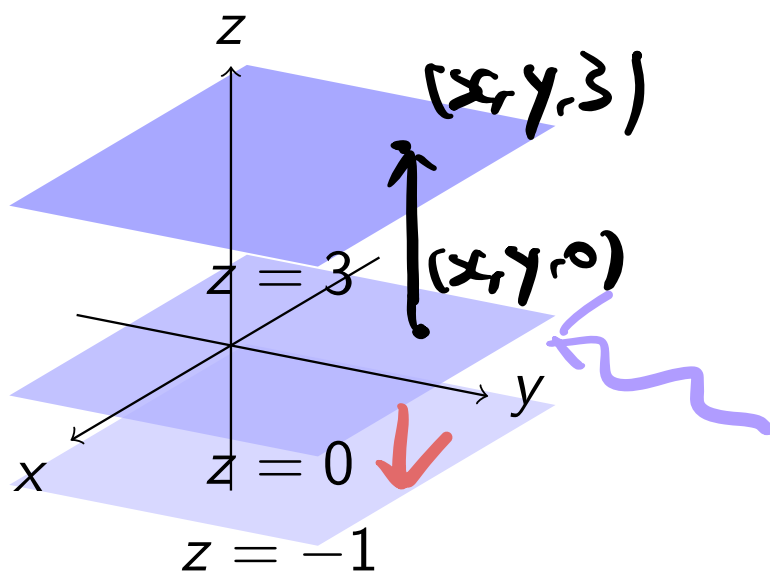
doesn't matter

$z=0$



Graphing equations in 3-space

The graph of an equation involving x , y , and z is the picture of all points (x, y, z) that satisfy the equation.

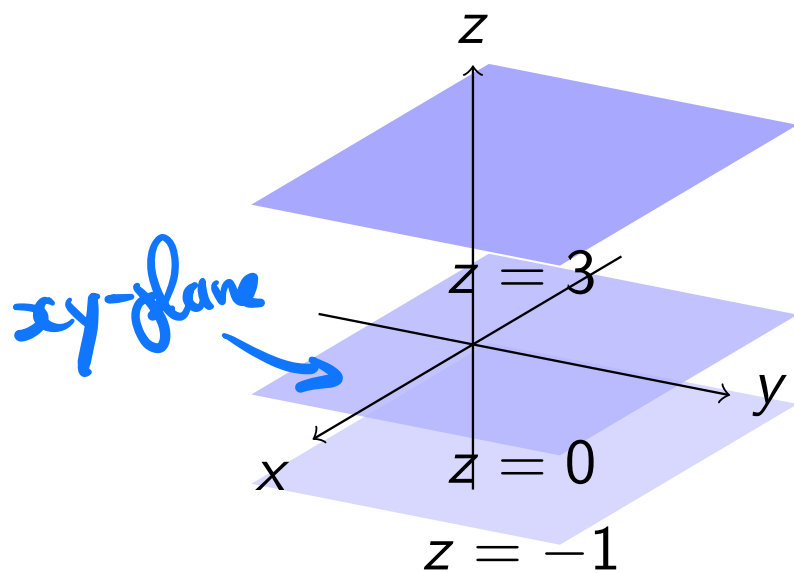


The graphs of $z = -1$, $z = 0$, and $z = 3$ are parallel planes.

$$\begin{aligned} z &= 0 \\ (x, y, z) &\text{ is a solution} \\ \Leftrightarrow z &= 0 \\ \{ (x, y, 0), \text{ } x, y &\text{ real numbers} \} \\ \hline z &= 3 \\ \{ (x, y, 3), x, y &\in \mathbb{R} \}. \end{aligned}$$

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The graph of an equation involving x , y , and z is the picture of all points (x, y, z) that satisfy the equation.



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Coordinate planes

There are three coordinate planes in 3-space:

xy -plane : $z = 0$,

xz -plane : $y = 0$,

yz -plane : $x = 0$.

3 axes, 3 coordinate planes

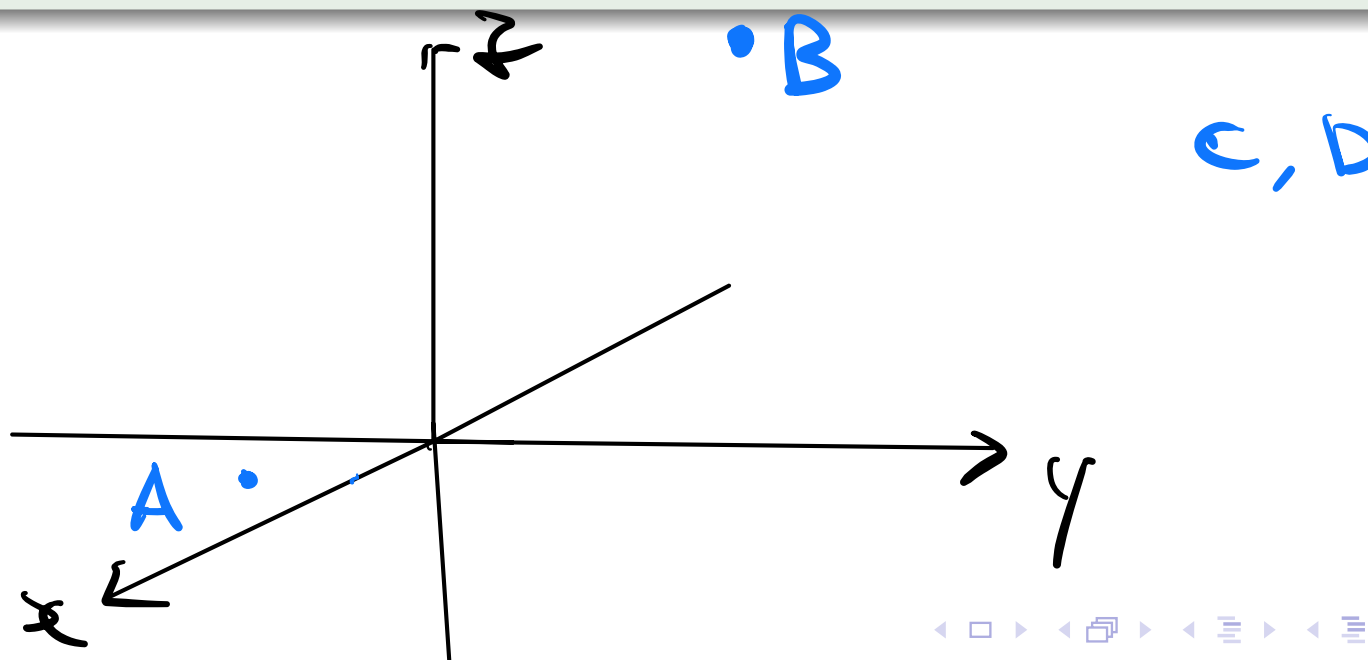
Points, planes, and axes

Example

Which of the points

$$A = (1, -1, 0), \quad B = (0, 3, 4), \quad C = (2, 2, 1), \quad D = (0, -4, 0)$$

lies closest to the xz -plane? Which point lies on the y -axis?



Points, planes, and axes

Example

Which of the points

desmos.com

→ y-axis

$A = (1, -1, 0)$, $B = (0, 3, 4)$, $C = (2, 2, 1)$, $D = (0, -4, 0)$

1 *3* *2* *4*

lies closest to the xz -plane? Which point lies on the y -axis?

Can you figure out the answers without drawing the points in 3-space?

How to quickly determine if a point lies in a coordinate plane or a coordinate axis?

*GOAL: algebraic way to locate these pts
10 minutes*

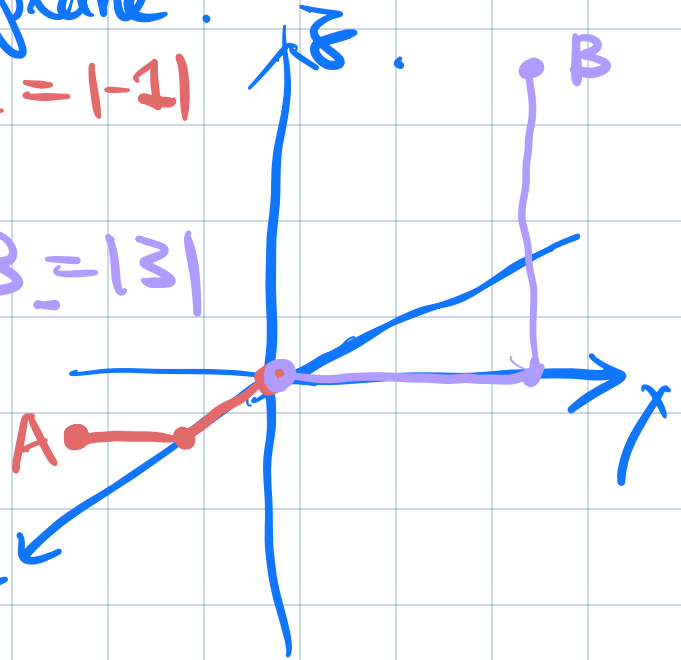
- (x, y, z)
- ① x units along x -axis
 - ② y units parallel to y -axis
 - ③ z units — — — —

Distance to xz -plane.

$A = (1, -1, 0)$ $0 \rightarrow 0 \rightarrow 1 \rightarrow 1 = |-1|$

$B = (0, 3, 4)$ $0 \rightarrow 0 \rightarrow 3 \rightarrow 3 = |3|$

The distance of
 (x, y, z) to xz -plane is
 $= |y|$



(x, y, z) lies on the y -axis.

$\Leftrightarrow x = z = 0$

• The distance of (x, y, z) to xy -plane
 $= |z|$

• (x, y, z) lies on the xy -plane $\Leftrightarrow z = 0$

• (x, y, z) lies on z -axis $\Leftrightarrow x = y = 0$

Describing sets of points

If a point has one coordinate equal to zero, it lies in a coordinate plane. If it has two coordinates equal to zero, it lies on a coordinate axis.

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If a point has one coordinate equal to zero, it lies in a coordinate plane. If it has two coordinates equal to zero, it lies on a coordinate axis.

Example

You are 2 units below the xy -plane and in the yz -plane. What are your coordinates?

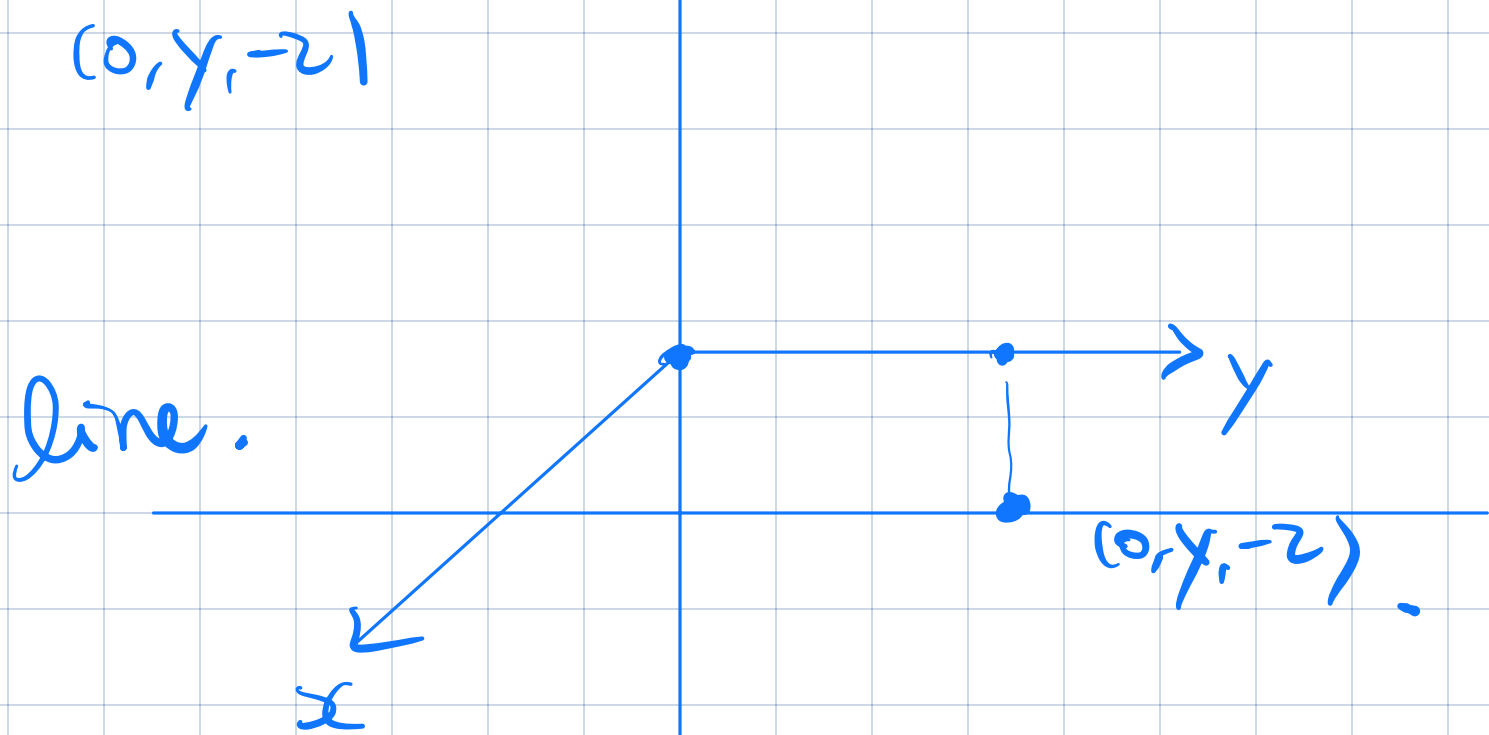
(x, y, z) is in yz -plane $\Leftrightarrow x=0$

$(0, y, z)$

its distance to xy -plane = $|z|$
"below"

$\Rightarrow |z| = 2 \Rightarrow z = -2$

$\Rightarrow (0, y, -2), y \in \mathbb{R}$



Describing sets of points

If a point has one coordinate equal to zero, it lies in a coordinate plane. If it has two coordinates equal to zero, it lies on a coordinate axis.

Example

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Example

What is behind the blackboard? Imagine that the yz -plane is our blackboard. Describe the region behind the blackboard algebraically.

$\hookrightarrow$ the positive x -axis points to our classroom

Distance between two points

(proof, ~~look~~)

Theorem

Distance formula in 3-space The distance between (x, y, z) and (a, b, c) is

$$\text{Distance} = \sqrt{(x - a)^2 + (y - b)^2 + (z - c)^2}.$$

2-dim space distance between
 (x, y) and (a, b)

$$= \sqrt{(x - a)^2 + (y - b)^2}$$

Distance between two points

Theorem

Distance formula in 3-space The distance between (x, y, z) and (a, b, c) is

$$\text{Distance} = \sqrt{(x - a)^2 + (y - b)^2 + (z - c)^2}.$$

Example

The distance between $(1, 2, 1)$ and $(-3, 1, 2)$ is

$$\sqrt{(-3 - 1)^2 + (1 - 2)^2 + (2 - 1)^2} = \sqrt{18}. = 3\sqrt{2}$$

A sphere and a solid ball

↪ represent it by equation.

Example

A sphere of radius 1 with center at the origin consists of all points (x, y, z) whose distance from the origin is 1.

The distance ^{between} (x, y, z) and $(0, 0, 0)$ is 1.

$$\Leftrightarrow \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2} = 1$$

$$\Leftrightarrow x^2 + y^2 + z^2 = 1$$

This sphere is the graph of q^n $x^2 + y^2 + z^2 = 1$



A sphere and a solid ball

Example

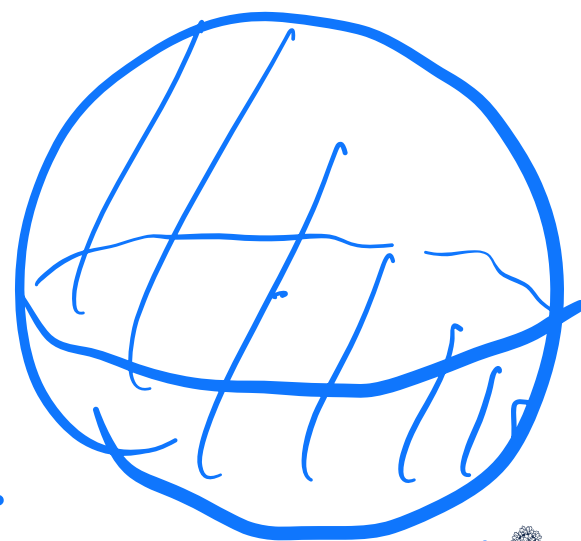
A sphere of radius 1 with center at the origin consists of all points (x, y, z) whose distance from the origin is 1.

The equation

$$x^2 + y^2 + z^2 = 1$$

represents the *surface of the sphere*.

what about (solid) inside?
 (x, y, z) is inside this sphere
 $\Leftrightarrow$ distance (x, y, z) and $0 \leq 1$
 $\Leftrightarrow \sqrt{x^2 + y^2 + z^2} \leq 1 \Leftrightarrow x^2 + y^2 + z^2 \leq 1$



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A sphere and a solid ball

A sphere of radius $r > 0$ w. center at (x_0, y_0, z_0) is represented by $(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = r^2$.

Example

A sphere of radius 1 with center at the origin consists of all points (x, y, z) whose distance from the origin is 1.

The equation

$$x^2 + y^2 + z^2 = 1$$

represents the *surface of the sphere*.

The solid ball enclosed by the sphere is represented by the inequality

$$x^2 + y^2 + z^2 \leq 1.$$

Intersection of the sphere

$$x^2 + y^2 + z^2 = 5$$

with the plane $z = 2$.

(x, y, z) lies in the intersection

$$\Leftrightarrow \begin{cases} x^2 + y^2 + z^2 = 5 \\ z = 2. \end{cases}$$

$$\Leftrightarrow \begin{matrix} (x, y, 2) \\ \text{c.t.} \end{matrix}$$

$$x^2 + y^2 = 5 - 2^2 = 1.$$

circle.

Summary

What we have learned this week:

- Describe multivariable functions.
- Visualize mathematical objects and regions in 3-dimensional space.
- Understand how to find the distance between two points.

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Spoilers for next week:

- How to describe the graph of a two-variable function in the 3-dimensional space?
- What can we read from the graph of a function?